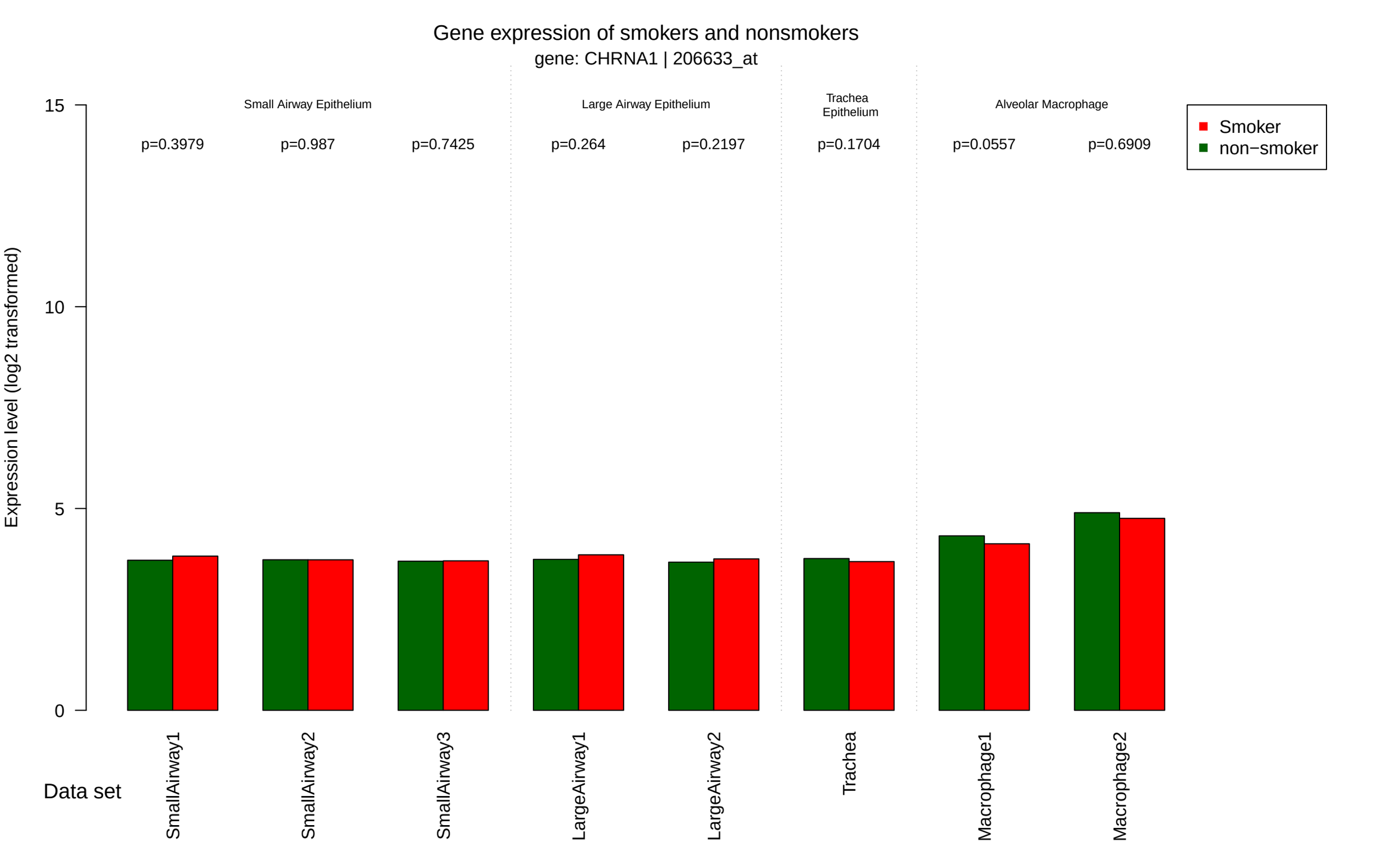
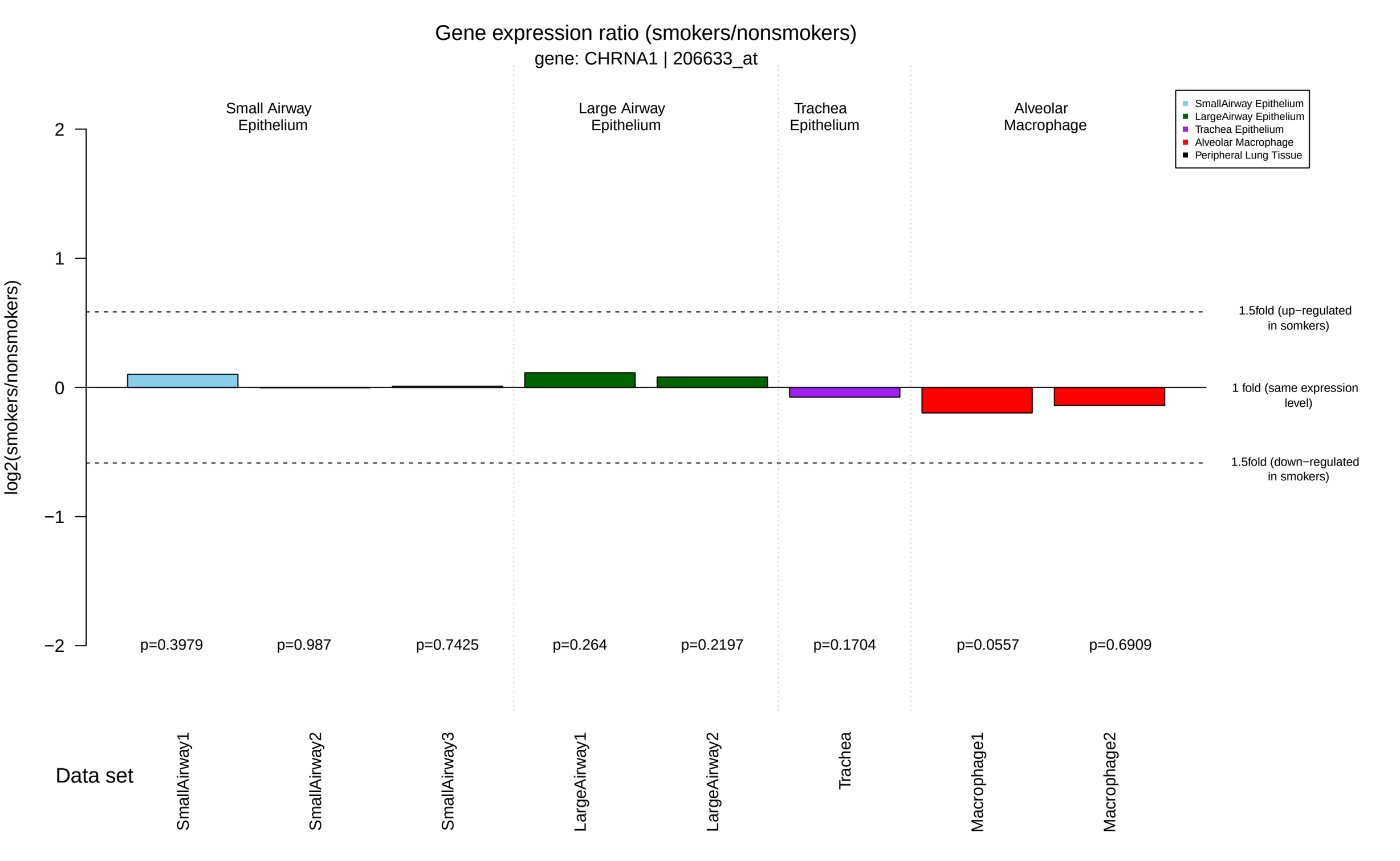
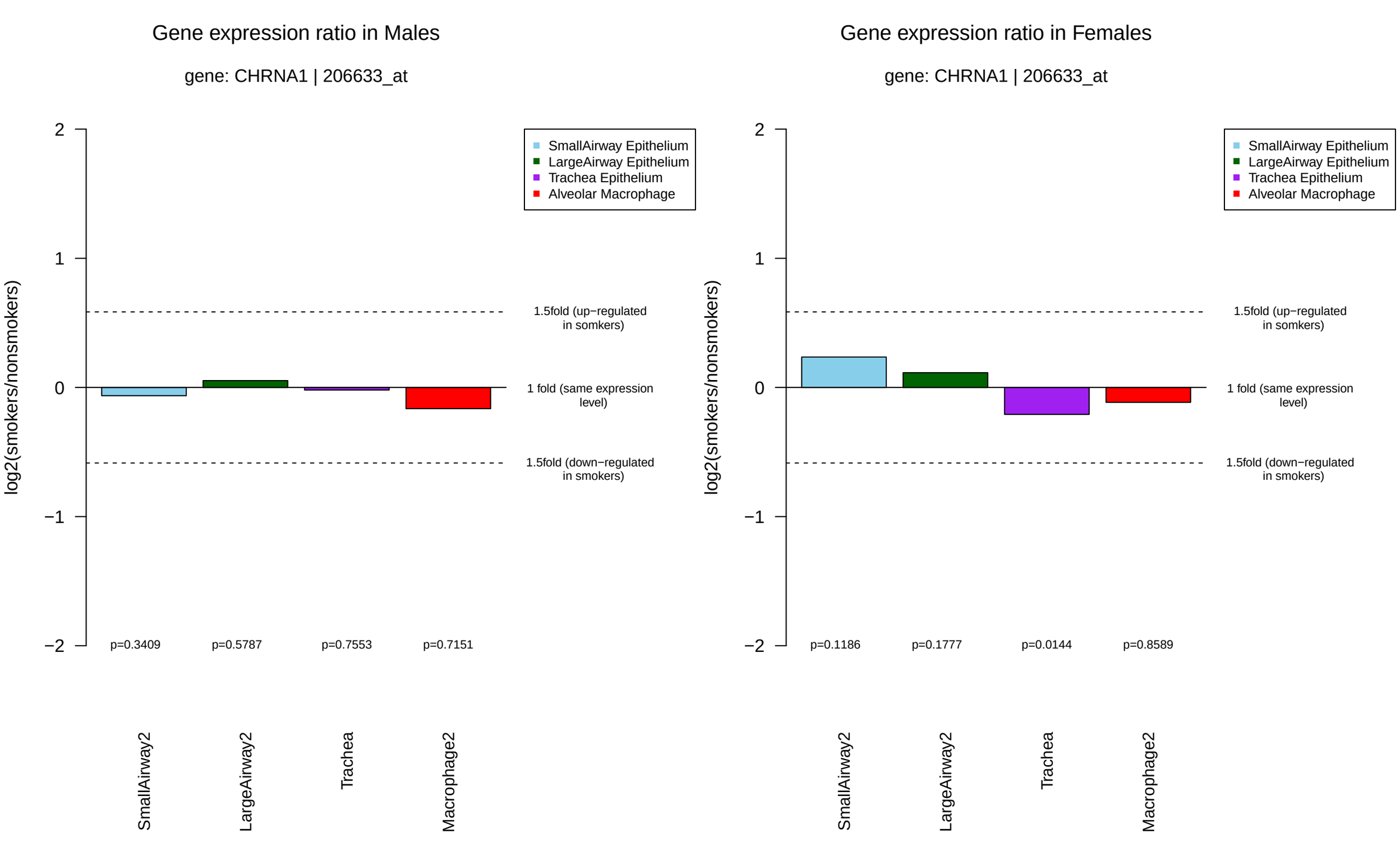
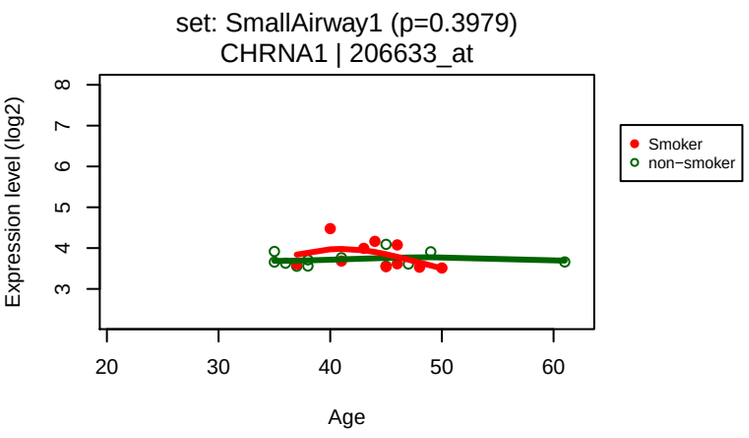
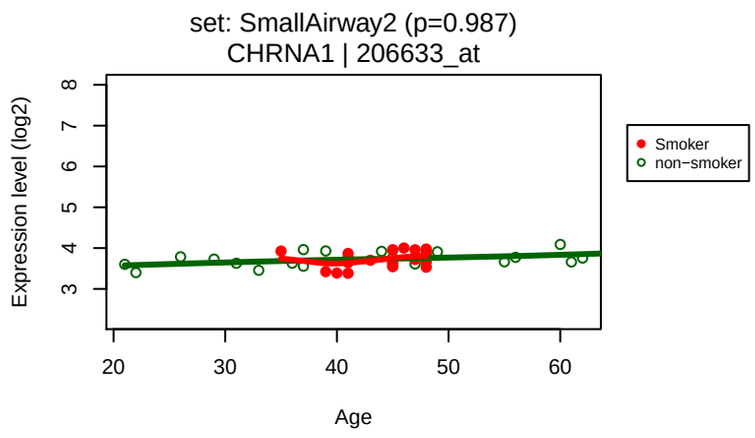
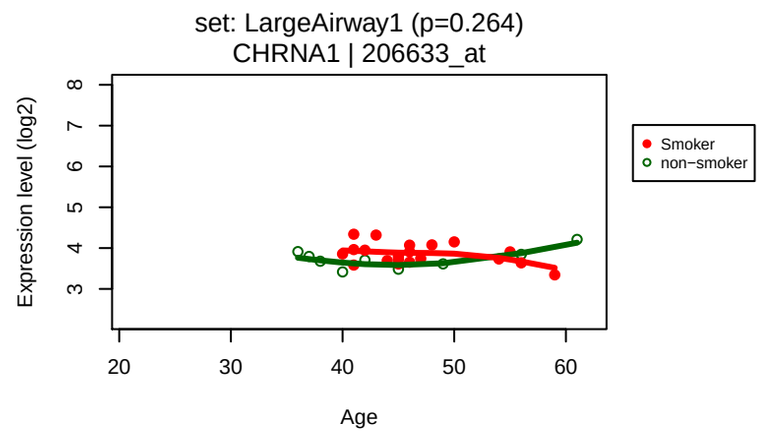

Gene expression of smokers and nonsmokers
gene: CHRNA1 | 206633_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3979
p=0.987
p=0.7425
p=0.264
p=0.2197
p=0.1704
p=0.0557
p=0.6909
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

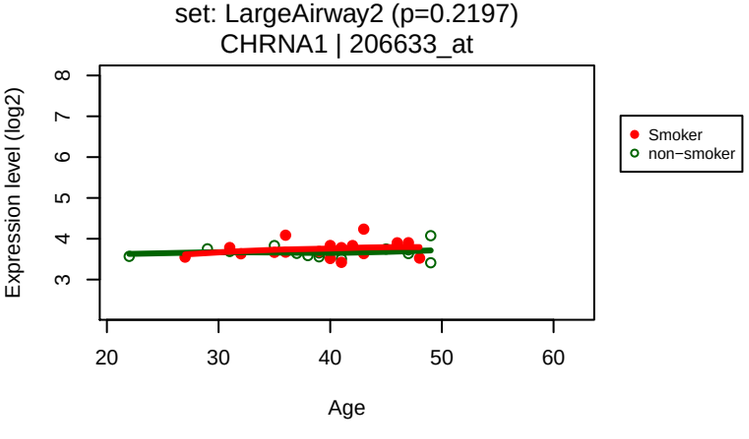

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA1 | 206633_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3979
p=0.987
p=0.7425
p=0.264
p=0.2197
p=0.1704
p=0.0557
p=0.6909
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

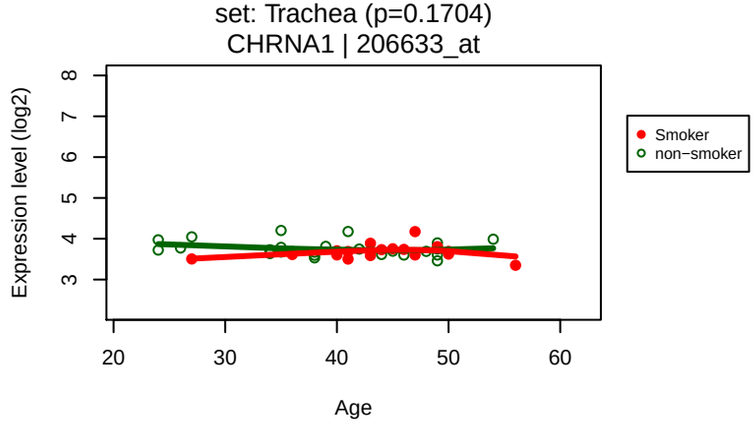

Gene expression ratio in Males
gene: CHRNA1 | 206633_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3409
p=0.5787
p=0.7553
p=0.7151
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA1 | 206633_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1186
p=0.1777
p=0.0144
p=0.8589
SmallAirway2
LargeAirway2
Trachea
Macrophage2

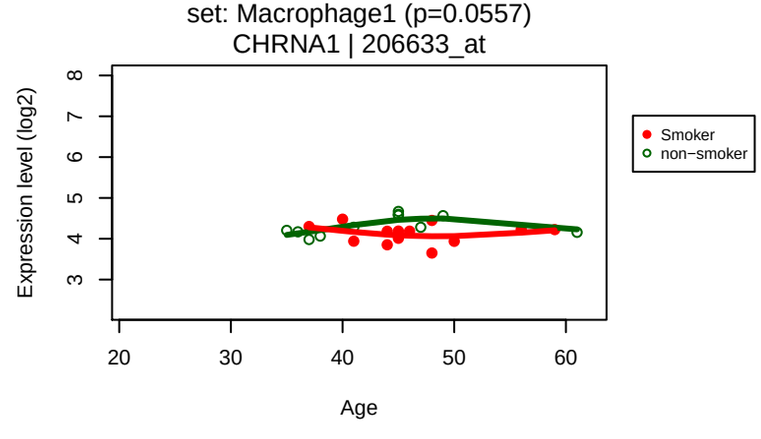

set: SmallAirway1 (p=0.3979)
CHRNA1 | 206633_at
Expression level (log2)
Age
Smoker
non−smoker

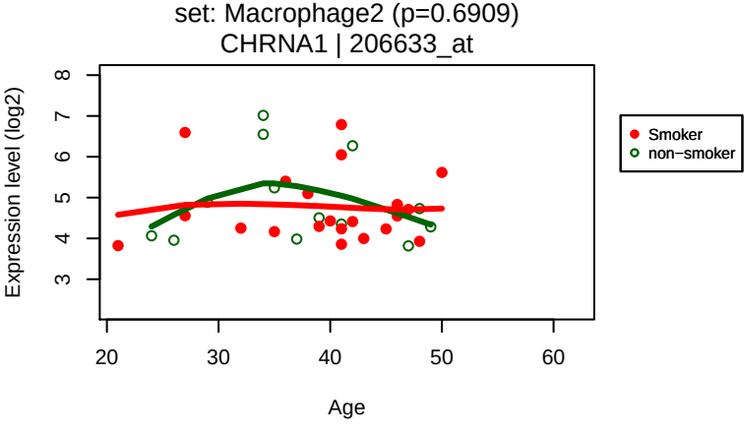

set: SmallAirway2 (p=0.987)
CHRNA1 | 206633_at
Expression level (log2)
Age
Smoker
non−smoker

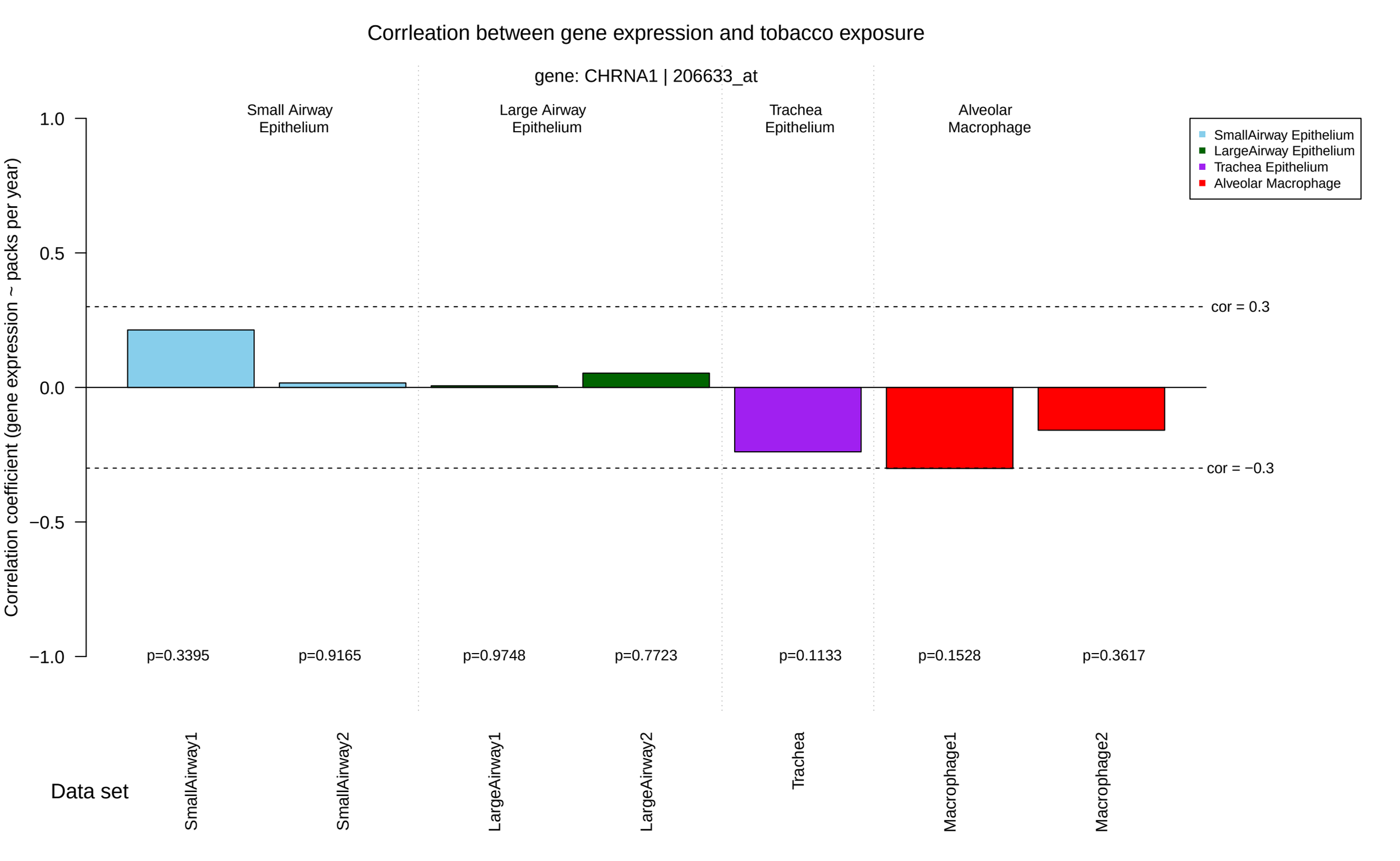

set: LargeAirway1 (p=0.264)
CHRNA1 | 206633_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2197)
CHRNA1 | 206633_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1704)
CHRNA1 | 206633_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0557)
CHRNA1 | 206633_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6909)
CHRNA1 | 206633_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA1 | 206633_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3395
p=0.9165
p=0.9748
p=0.7723
p=0.1133
p=0.1528
p=0.3617
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2